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FORM OWNED TO PETROBRAS N-0381 REV.L.

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1 INTRODUCTION

1.1 OBJECT

1.1.1 This specification describes the minimum requirements for the adequate interfacing of the Main Packages' Automation and Instrumentation System with the UNIT.

1.1.2 This technical specification deals exclusively with the interfaces related to Automation and Instrumentation. For interface with the other systems, see respective discipline project documentation.

1.2 DEFINITIONS

1.2.1 Refer to I-ET-3010.00-1200-940-P4X-002 - GENERAL TECHNICAL TERMS.

1.3 ABBREVIATIONS, ACRONYMS AND INITIALISMS

AEPR	Automation & Electrical Panels Room
CCR-ATR	Central Control Room – Automation and Turbomachinery Room
CSS	Control and Safety System
DIO	Optical Internal Distributor
ESD	Emergency shutdown
FAT	Factory Acceptance Test
I/O	Input / Output
LAN	Local Area Network
MMS	Machinery Monitoring System
MPA	Automatized Procedures Module (*Portuguese*: Módulo de Procedimentos Automatizados)
MPS	Machinery Protection System
OPC	Open Platform Communications
PAS	Package Automation System
PCS	Process Control System
PLC	Programmable Logic Controller
PSD	Process Shutdown System
RESD	Emergency Shutdown Relay
RIO	Remote I/O
SAT	Site Acceptance Test
SIT	Site Integration Test
SOS	Supervision and Operation System
TAP	Performance Acceptance Test
TCP/IP	Transmission Control Protocol/Internet Protocol
UCP	PACKAGE UNIT Control Panel
Vac	Unit for AC voltage
Vdc	Unit for DC voltage

2 REFERENCE DOCUMENTS, CODES AND STANDARDS

2.1 EXTERNAL REFERENCES

2.1.1 International Codes, Recommended Practices and Standards

IEC - INTERNATIONAL ELECTROTECHNICAL COMMISSION

IEC 60079	EXPLOSIVE ATMOSPHERES – ALL PARTS
IEC 60092-350	ELECTRICAL INSTALLATIONS IN SHIPS – PART 350 - GENERAL CONSTRUCTION AND TEST METHODS OF POWER, CONTROL AND INSTRUMENTATION CABLES FOR SHIPBOARD AND OFFSHORE APPLICATIONS
IEC 60092-376	ELECTRICAL INSTALLATIONS IN SHIPS – PART 376 - CABLES FOR CONTROL AND INSTRUMENTATION CIRCUITS 150/250 V (300 V)
IEC 60079-25	EXPLOSIVE ATMOSPHERES – PART 25: INTRINSICALLY SAFE ELECTRICAL SYSTEMS
IEC 60092-504	ELECTRICAL INSTALLATIONS IN SHIPS - PART 504: SPECIAL FEATURES - CONTROL AND INSTRUMENTATION
IEC 60529	DEGREE OF PROTECTION (IP CODE)
IEC 60533	ELECTRICAL AND ELECTRONIC INSTALLATIONS IN SHIPS - ELECTROMAGNETIC COMPATIBILITY (EMC) – SHIPS WITH METALLIC HULL
IEC 61000	ELECTROMAGNETIC COMPATIBILITY (EMC) SERIES - ALL PARTS
IEC 61280	FIBRE OPTIC COMMUNICATION SUBSYSTEM BASIC TEST PROCEDURES – ALL PARTS
IEC 62337	COMMISSIONING OF ELECTRICAL, INSTRUMENTATION AND CONTROL SYSTEMS IN THE PROCESS INDUSTRY – SPECIFIC PHASES AND MILESTONES
IEC 62381	AUTOMATION SYSTEMS IN THE PROCESS INDUSTRY - FACTORY ACCEPTANCE TEST (FAT), SITE ACCEPTANCE TEST (SAT) AND SITE INTEGRATION TEST (SIT)

2.1.1.1 All *Secretaria de Inspeção do Trabalho* Regulatory Standards (NRs) shall be followed.

2.1.2 Classification Society

2.1.2.1 The DETAIL DESIGN PHASE shall be submitted to approval by Classification Society. The design and installation shall take into account their requirements and comments.

2.1.2.2 The design, installation and operation shall strictly follow the classification society requirements, along with the specific requirements identified in this document, including also all referenced documents' requirements.

2.2 INTERNAL REFERENCES

DR-ENGP-M-I-1.3	SAFETY ENGINEERING
I-ET-3010.00-1200-800-P4X-002	AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS
I-ET-3010.00-1200-940-P4X-002	GENERAL TECHNICAL TERMS
I-ET-3010.1M-1200-800-P4X-005	FIELD INSTRUMENTATION
I-ET-3010.00-5520-800-P4X-001	SUPERVISION AND OPERATION SYSTEM (SOS) SCREENS
I-ET-3010.00-5520-888-P4X-001	AUTOMATION PANELS
I-ET-3010.00-5140-700-P4X-003	ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS
I-ET-3010.00-1200-850-P4X-002	ASSET MANAGEMENT SYSTEM (AMS)
I-ET-3010.00-5500-854-P4X-001	MACHINERY MONITORING SYSTEM (MMS)
I-ET-3010.1M-1225-323-P4X-001	VAPOR RECOVERY UNIT
I-ET-3010.1M-1231-321-P4X-001	MAIN GAS MOTOCOMPRESSOR
I-ET-3010.1M-1231-321-P4X-002	EXPORTATION GAS MOTOCOMPRESSOR
I-ET-3010.1M-1233-560-P4X-002	MOLECULAR SIEVE UNIT
I-ET-3010.1M-1235-560-P4X-001	CO2 REMOVAL UNIT
I-ET-3010.1M-1252-321-P4X-001	INJECTION GAS MOTOCOMPRESSOR
I-ET-3010.1M-1254-321-P4X-001	CO2 TURBOCOMPRESSOR
I-ET-3010.1M-5147-332-P4X-001	TECHNICAL SPECIFICATION FOR TURBOGENERATOR PACKAGE
I-DE-3010.1M-5520-800-P4X-004	NETWORK INTERCONNECTION DIAGRAM
I-ET-3010.00-1200-800-P4X-010	CRITERIA FOR ESTABLISHING CABLE CODES AND CABLE GLAND CODES
I-ET-3010.00-1200-800-P4X-013	GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS
I-DE-3010.1M-1200-942-P4X-002	GENERAL ARRANGEMENT
I-DE-3010.1M-5520-800-P4X-002	AUTOMATION AND CONTROL ARCHITECTURE
I-ET-3010.1M-5520-800-P4X-004	AUTOMATION NETWORK REQUIREMENTS
I-ET-3010.1M-1200-800-P4X-001	INSTRUMENTATION ADDITIONAL TECHNICAL REQUIREMENTS
I-MD-3010.1M-5520-800-P4X-003	AUTOMATION NETWORK DESCRIPTION
I-ET-3010.00-5139-390-P4X-001	HYDRAULIC POWER UNIT (HPU) FOR TOPSIDES VALVES.
I-DE-3010.1M-5400-94A-P4X-001	AREA CLASSIFICATION - GENERAL
I-RL-3010.1M-1200-940-P4X-001	GENERAL SPECIFICATION FOR AVAILABLE UTILITIES.

2.2.1 When more restrictive, Brazilian regulation (MTE section) and INMETRO regulation superpose all codes and regulations listed in item 2.2, since they are enforced by Brazilian law.

3 GENERAL – INFORMATION COMMON TO ALL PACKAGES

3.1 GENERAL REQUIREMENTS

3.1.1 For each package type requirements and definitions, refer to I-ET-3010.00-1200-800-P4X-002 – AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS

3.1.2 During detailed engineering design, it shall be issued four additional documents regarding cables:

- Cable List;
- Cable Code list;
- Cable Gland Code List;
- Cable tray occupation calculation memory.

3.1.2.1 Document I-ET-3010.00-1200-800-P4X-010 – CRITERIA FOR ESTABLISHING CABLE CODES AND CABLE GLAND CODES, contains the criteria for emitting such documents.

3.1.3 Redundant network cables shall have segregated routing in such a way that a cause of damage to one of the cables will not affect the other.

3.1.4 UCP shall be installed at the Automation and Electrical Panels Room (AEPR), Central Control Room – Equipment Ambiance (CCR-EA) or at field (where designated) while RIO Panels (where applicable, as in item 3.1.10) shall be installed at field.

3.1.5 For UCPs and RIOs locations, see I-LI-3010.1M-1200-940-P4X-002 – EQUIPMENT LIST.

3.1.6 All PACKAGE UNIT Fire and Gas signals shall be sent to the UCP.

3.1.7 For operational and environmental conditions additional to this section, see I-ET-3010.1M-1200-800-P4X-001 - INSTRUMENTATION ADDITIONAL TECHNICAL REQUIREMENTS.

3.1.8 The available power supply for UCP and RIO panels shall be in accordance with I-ET-3010.00-5140-700-P4X-003 - ELECTRICAL REQUIREMENTS FOR PACKAGES FOR OFFSHORE UNITS. The available power to be supplied by panels for instrumentation shall be 24 Vdc unless otherwise specified on I-ET-3010.1M-1200-800-P4X-005 - FIELD INSTRUMENTATION.

3.1.9 All panels, materials, instruments and equipment proper to be used in hazardous areas, shall have conformity certificates according to I-ET-3010.1M-1200-800-P4X-001 - INSTRUMENTATION ADDITIONAL TECHNICAL REQUIREMENTS. All electrical and instrumentation equipment and controls installed in hazardous areas, according to I-DE-3010.1M-5400-94A-P4X-001 – AREA CLASSIFICATION - GENERAL shall be certified for installation in hazardous areas appropriate zone classification. All electrical and instrumentation equipment and controls installed in external area outside hazardous areas that are kept turned on during gas leakage, i.e., ESD 3, shall be certified for installation in hazardous areas as well, certified for Zone 2. All INMETRO certificates for these equipment shall be provided before FAT for PETROBRAS approval.

3.1.10 Main, Exportation, CO2 and Injection Compressors and Vapor Recovery Unit shall have a package RIO Panel each. For more information on UCP and RIO Panel specifications and required functionalities, see its respective PACKAGE TECHNICAL SPECIFICATION.

4 INTERFACE SIGNALS HARDWIRED TO CSS

4.1 SIGNALS

4.1.1 The minimal interface signals between PACKAGE UNITs and CSS are according annex spreadsheet, below. Other signals may be necessary during Detail Engineering Design.

Interface signals.xlsx

4.1.2 Whenever a signal is mentioned for a package that has multiplicity, such as a compression unit composed of 3 compression trains, UC-1231001A/C for example, the signal is applicable to each one of the identical packages. For example, the signal for the aforementioned case should be XSL-1231001A, XSL-1231001B and XSL-1231001C – one per identical PACKAGE.

5 INTERFACE SIGNALS SENT VIA NETWORK TO CSS

5.1 SIGNALS

5.1.1 All P2, P2S, P2C and P2SC PACKAGE UNIT shall send through the network interface, at least, the following signals:

- All process variables (including totalizers);
- All valve statuses (limit switch, position indicator, commands to valves);
- All alarm set points;
- All alarm statuses;
- All controller set points (including PID internal variables like Kp, Ki, Kd and internal accumulators);
- All equipment (pumps, compressors etc.) statuses (Running/Stopped);
- All equipment (pumps, compressors etc.) VSD set point (for those with VSD);
- All equipment (pumps, compressors etc.) stepped capacity set point (for those with stepped capacity set point);
- Timers for batch processes cycles (ex.: adsorption cycles, membrane cycle etc.).

5.1.2 In order to standardize variables and alarms presentation for automation integration services and historical recording in SOS, the following variables to be presented in communication map of Package unit´s controller, to be read by SOS Package Units Data Servers, shall have the values below:

- Variable in Normal Condition = 0;
- Variable in Abnormal Condition = 1;
- Logical outputs: Engine ON (Running), Valve Opened (all valves) = 1
- Non-active alarm: 0; Active alarm: 1;

- Not acknowledged Alarm: 0, Acknowledged alarm: 1
- Levels: normalized to 0 – 100 %;
  NOTE: Since 1 means “valve opened”, the output of the BDV and fail open XV blocks shall be connected to a logical inverter (“0” to “1” and “1” to “0”).

6 FIELD INSTRUMENTS INTERFACE

6.1 INTERFACE

6.1.1 All skid instruments shall be connected to the specific junction box on the skid according to I-ET-3010.00-1200-800-P4X-013 - GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS and PACKAGE TECHNICAL SPECIFICATION.

6.1.2 At least one interface junction box shall be provided on the module for each skid. These junction boxes shall be specified and segregated according to I-ET-3010.00-1200-800-P4X-013 - GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS.

6.1.3 All signals provided by instruments on process plant related to anti-surge and capacity control will be connected by multicables to the corresponding module’s interface junction boxes and from these junction boxes to the respective UCPs on AEPR.

6.1.4 Never, in any case, BDVs shall be used to other purpose than emergency depressurization (such as process or commissioning depressurization, in which case a XV shall be used).

6.1.5 All cables connecting to the BDV and SDV solenoid valves and limit switch shall be fire resistant (see I-ET-3010.00-1200-800-P4X-013 - GENERAL CRITERIA FOR INSTRUMENTATION PROJECTS).

7 NETWORK INTERFACE

7.1 GENERAL

7.1.1 PETROBRAS will inform, during project’s detailing design phase, a list of available IP addresses to be used for the connection between the PACKAGE UNIT and PACKAGE UNIT LAN, according to I-MD-3010.1M-5520-800-P4X-003 - AUTOMATION NETWORK DESCRIPTION and I-ET-3010.00-1200-800-P4X-002 - AUTOMATION, CONTROL AND INSTRUMENTATION ON PACKAGE UNITS.

7.1.2 Each UCP of the PACKAGE UNIT shall be connected to PACKAGE UNIT LAN at panel PN-5523001 (TOPSIDES SOS SERVERS PANEL, in case of a Topsides PACKAGE) or to panel PN-5523501 (HULL SOS SERVERS PANEL, in case of a Hull PACKAGE) through a redundant Gigabit Ethernet link, according to I-DE-3010.1M-5520-800-P4X-004 – NETWORK INTERCONNECTION DIAGRAM and I-ET-3010.1M-5520-800-P4X-004 – AUTOMATION NETWORK REQUIREMENTS.

7.1.3 Ethernet connections between PACKAGE UNIT and SOS shall be according to I-ET-3010.1M-5520-800-P4X-004 – AUTOMATION NETWORK REQUIREMENTS.

7.1.4 It shall be provided UCP controllers with 2 OPC UA server drivers. These OPC drivers shall be installed in PACKAGE UNIT DATA SERVERS, in order to standardize the communication between UCP and PACKAGE UNIT DATA SERVERS. The OPC driver shall be OPC Foundation™ compliant. Only in case the UCP doesn't support OPC Server driver, PETROBRAS accepts Modbus/TCP driver or a dedicated communication driver for communication between UCP and PACKAGE UNIT DATA SERVERS. This link shall be used to send all variables described in item 5.1.1 to CSS. A simplified version of the PACKAGE UNIT operation screen shall be made available in SOS HMIs using these variables.

7.2 AMS INTERFACE

7.2.1 AMS System shall acquire data from PACKAGE UNIT through the PACKAGE UNIT LAN. For more information on AMS System, see I-ET-3010.00-1200-850-P4X-002 – ASSET MANAGEMENT SYSTEM (AMS).

7.3 MMS INTERFACE

7.3.1 For P0 PACKAGE UNITS (see appendix for PACKAGE UNIT classification), the machinery monitoring signals will be connected to Hull Protection and Acquisition Panel (PN-5500506), in case of a Hull Machine, or Topsides Protection and Acquisition Panel (PN-5500009), in case of a Topsides Machine.

7.3.2 For P2/P2S/P2C/P2SC PACKAGE UNITS (see appendix for PACKAGE UNIT classification), the UCP for PACKAGE UNITS (or RIO Panel, where it exists – see item 3.1.10) shall provide connection from MPS inside UCP Panel (or RIO Panel) to MMS Server Panel (PN-5500007/PN-5500014). This connection will be made through redundant communications. If necessary, conversion between fiber optic and twisted pair shall be performed in a converter mounted inside DIOs panel.

7.3.3 For more information on MMS System, see I-ET-3010.00-5500-854-P4X-001 – MACHINERY MONITORING SYSTEM (MMS).

7.4 HMI INTERFACE

7.4.1 All P2S (according to classification on appendix I) UCPs will be connected to a network switch in the AEPR that will be connected to its HMI at the CCR-OA, such as defined in I-DE-3010.1M-5520-800-P4X-002 - AUTOMATION AND CONTROL ARCHITECTURE.

7.4.2 For the main, exportation, CO2 and injection compressors, the aforementioned switch will be connected to the workstations PN-5500010A/D (COMPRESSOR CAPACITY CONTROL SYSTEM WORKSTATIONS) at the CCR-OA, such as defined in I-DE-3010.1M-5520-800-P4X-002 - AUTOMATION AND CONTROL ARCHITECTURE.

7.4.3 All P2 PACKAGE UNITS, that do not have a dedicated HMI in CCR-OA, shall be able to have its UCP internal HMI accessed via PN-5523006A/B – REMOTE ACCESS WORKSTATION. These workstations will be located in CCR-OA and shall be able to access the UCPs internal HMI via Windows Remote Desktop. If additional licenses or passwords are necessary to perform this access, they shall

be supplied along with the PACKAGE UNIT. This HMI CPU shall be suitable for mounting a 19” rack.

7.5 UCP AND RIO PANEL INTERFACE

7.5.1 Where applicable (see item 3.1.10) Control RIO, Safety RIO and Overspeed Protection shall be connected to a DIOs PANEL by a redundant optical fiber network connection and from the aforementioned DIOs Panel to UCP by twisted pair network cables. These communication protocols may be proprietary.

7.5.2 This RIO panel shall be certified as Ex-pz, certified for Zone 2, according to IEC-60079. An air vessel shal be supplied to keep the panel protected in case of loss of air from the FPU. Vessel sizing shall be submitted to PETROBRAS. For further details, see I-ET-3010.00-5520-888-P4X-001 - AUTOMATION PANELS.

7.5.3 For Main, Exportation, CO2 and Injection Compressors and Vapor Recovery Unit, the internal components of the UCP and RIO are depicted in figure 1, below.

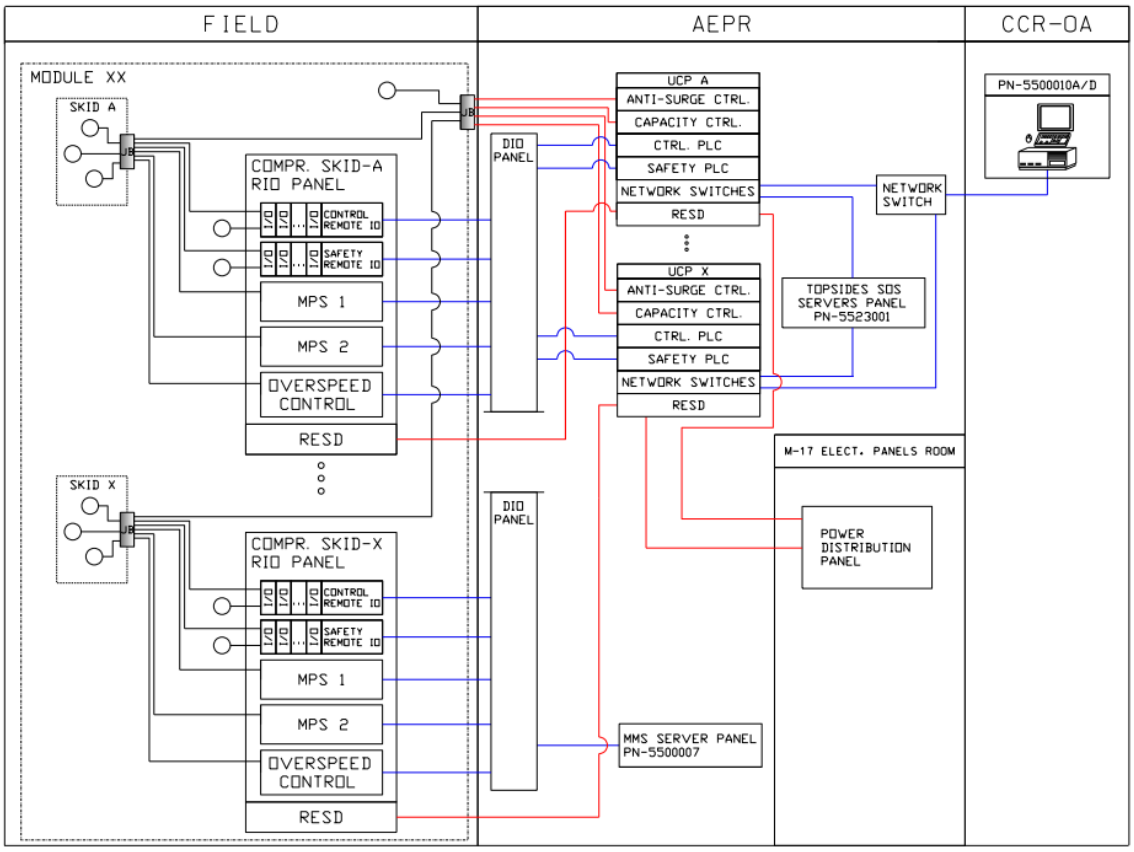

Figure 1 – Internal Components

7.6 COMPRESSOR CONTROL MPA INTERFACE

7.6.1 Main, Exportation, CO2 and injection compressors and Vapor Recovery Unit will have an interface with MPA software to perform advanced control. This MPA software will be installed at BR Optimization Server.

7.6.2 All signals necessary to MPA software will be sent through network interface from PACKAGE UNIT UCP and PACKAGE UNIT RTDS. These signals shall be available in the PACKAGE UNIT communication map. For more information on these signals, see item 5.1.1.

7.6.3 The MPA software will read these data from PACKAGE UNIT RTDS through SOS LAN. The result of the MPA control will then be sent to PCS via network (CSS LAN), and then to the PACKAGE UNIT UCP via hardwired analog signals. For more information on these signals, see 4.1.1.

7.6.4 Since these signals will command loops from the PACKAGE UNIT UCP, there shall be a selection logic between the signal from the PACKAGE UNIT itself and the signal from MPA software. This selection logic shall be done in PACKAGE UNIT UCP. For more details, see I-ET-3010.1M-1225-323-P4X-001 - VAPOR RECOVERY UNIT, I-ET-3010.1M-1231-321-P4X-001 - MAIN GAS MOTOCOMPRESSOR, I-ET-3010.1M-1231-321-P4X-002 - EXPORTATION GAS MOTOCOMPRESSOR, I-ET-3010.1M-1252-321-P4X-001 - INJECTION GAS MOTOCOMPRESSOR and I-ET-3010.1M-1254-321-P4X-001 - CO2 TURBOCOMPRESSOR.

7.6.5 The aforementioned communications are depicted in figure 2, below:

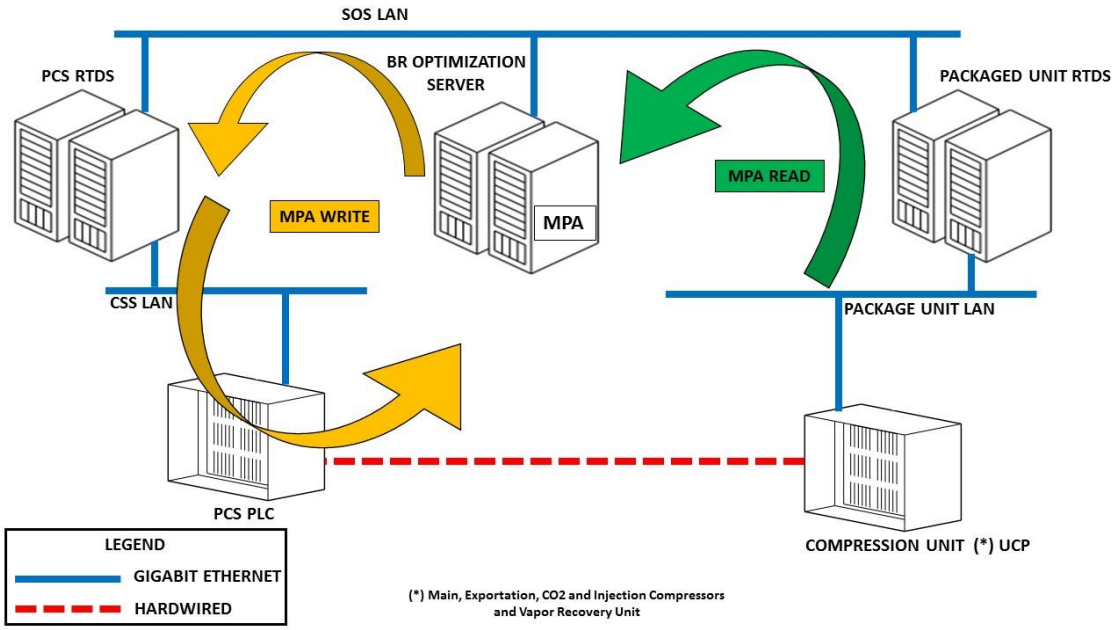

Figure 2 – MPA Communications

8 PACKAGE RIO PANEL AND UCP HARDWIRED INTERFACE

8.1 INTERFACE

8.1.1 For PACKAGE UNITS where such panel exists (see item 3.1.10), there shall be a hardwired connection between the Emergency Shutdown Relay (RESD) installed on RIO Panel and the one installed on UCP. For more information regarding the RESD and its functionality, see PACKAGE TECHNICAL SPECIFICATION.

9 PNEUMATIC/HYDRAULIC INTERFACE

9.1 INTERFACE

9.1.1 Essential air supply shall be made available by the UNIT to the PACKAGE UNITS that need it and to the RIO Panels (where such panel exists, see item 3.1.10). For details about specifications for service air, see I-ET-3010.1M-1200-800-P4X-001 - INSTRUMENTATION ADDITIONAL TECHNICAL REQUIREMENTS and I-RL-3010.1M-1200-940-P4X-001 - GENERAL SPECIFICATION FOR AVAILABLE UTILITIES.

9.1.2 In case a Topsides PACKAGE UNIT needs hydraulic fluid to power an internal component, PETROBRAS shall be consulted beforehand. For details about specifications for hydraulic fluid, see I-ET-3010.00-5139-390-P4X-001 - HYDRAULIC POWER UNIT (HPU) FOR TOPSIDES VALVES.

9.1.3 Service air tubings diameters and hydraulic fluid tubings diameters shall be defined during Detailing Engineering Design.

I - APPENDIX I – PACKAGE CLASSIFICATION

The classification of each package according to the definition from I-ET-3010.00-1200-800-P4X-002 – AUTOMATION, CONTROL AND INSTRUMENATION ON PACKAGE UNITS. Packages not specifically defined in this appendix are considered as P0.

PACKAGE TAG	PACKAGE NAME	TYPE	UCP TAG	LOCATION
AC-5252503A/B	FWD PANELS ROOM SELFCONTAINED UNIT	P1	PN-AC-5252503A/B	HULL
AC-5252619	SELF-CONTAINED UNIT FOR CCR AUTOM. & TBM ROOM / SUPROD / CCR OPER. AMB / EMERG. RESP. CENTER	P1	PN-AC-5252619	HULL
AC-5252620	SEL-CONTAINED UNIT FOR TELECON CCR / CCR EQUIP. AMBIENCE (AIR HANDLING)	P1	PN-AC-5252620	HULL
AC-5252623	SELF-CONTAINED UNIT FOR TELECOM CONTROL ROOM/ TELECOM PANEL ROOM	P1	PN-AC-5252623	HULL
B-1251001A/C	BOOSTER INJECTION WATER PUMP	P0	---	TOPSIDES
B-1251002A/C	MAIN INJECTION WATER PUMP	P2	PN-B-1251002A/C	TOPSIDES
B-5111001A/G	SEA WATER LIFT PUMP	P2	PN-B-5111001A/G	TOPSIDES
B-5111002	EMERGENCY SEA WATER LIFT PUMP	P2	PN-B-5111002	TOPSIDES
B-5124001A/C	COOLING WATER CIRCULATION PUMP - CLASSIFIED AREA	P0	---	TOPSIDES
B-5124002A/C	COOLING WATER CIRCULATION PUMP - NON-CLASSIFIED AREA	P0	---	TOPSIDES
B-5124003	EMERGENCY COOLING WATER PUMP - NON-CLASSIFIED AREA	P0	---	TOPSIDES
B-5125001A/C	HOT WATER CIRCULATION PUMP	P0	---	TOPSIDES
B-5125003A/B	HOT WATER DUMP COOLER PUMP	P0	---	TOPSIDES
B-5133001A/B	WELL SERVICE PUMP	P1	PN-B-5133001A/B	TOPSIDES

PACKAGE TAG	PACKAGE NAME	TYPE	UCP TAG	LOCATION
B-5133003	LEAK TEST PUMP	P1	PN-B-5133003	TOPSIDES
B-UB-5420501A/C-01	FIRE WATER LIFT PUMP	P1	PN-B-UB-5420501A/C-01	HULL
B-UB-5420501A/C-02	FIRE WATER BOOSTER PUMP	P0	---	HULL
B-Z-5424501-A/B	FOAM CONCENTRATE PUMP	P1	PN-B-Z-5424501A/B	HULL
CJ-5139501A/F	CHAIN STOPPER FWD PORTSIDE	P2	PN-CJ-5139501A/F	HULL
CJ-5139503A/F	CHAIN STOPPER FWD STARBOARD	P2	PN-CJ-5139503A/F	HULL
CJ-5139504A/F	CHAIN STOPPER AFT PORTSIDE	P2	PN-CJ-5139504A/F	HULL
CJ-5139505A/F	CHAIN STOPPER AFT STARBOARD	P2	PN-CJ-5139505A/F	HULL
FL-5331001A/B	GAS FLOTATION UNIT	P0	---	TOPSIDES
GD-5266501	FIXED BOOM CRANE	P1	PN-GD-5266501	HULL
GD-5266502	FIXED BOOM CRANE	P1	PN-GD-5266502	HULL
GG-5241501A/B	INERT GAS GENERATOR	P2S	PN-GG-5241501A/B-02	HULL
GN-Z-1359502-01	AFT OFFLOADING HOSE WINCH & REEL	P1	PN-GN-Z-1359502-01	HULL
GN-Z-1359502-02	AFT OFFLOADING HAWSER WINCH & REEL	P1	PN-GN-Z-1359502-02	HULL
GN-Z-1359506-01	FWD OFFLOADING HOSE WINCH & REEL	P1	PN-GN-Z-1359506-01	HULL
GN-Z-1359506-02	FWD OFFLOADING HAWSER WINCH & REEL	P1	PN-GN-Z-1359506-02	HULL
PN-1210001A/B	SUBSEA MASTER CONTROL STATION	P2SC	PN-1210001A/B	TOPSIDES
PN-1210002A/B	SUBSEA MASTER CONTROL STATION	P2SC	PN-1210002A/B	TOPSIDES
PN-1210003A/B	SUBSEA MASTER CONTROL STATION	P2SC	PN-1210003A/B	TOPSIDES
PN-1210004A/B	SUBSEA MASTER CONTROL STATION	P2SC	PN-1210004A/B	TOPSIDES
PN-1210005A/B	SUBSEA MASTER CONTROL STATION	P2SC	PN-1210005A/B	TOPSIDES
PN-1210006A/B	SUBSEA MASTER CONTROL STATION	P2SC	PN-1210006A/B	TOPSIDES
PN-1210007A/B	SUBSEA MASTER CONTROL STATION	P2SC	PN-1210007A/B	TOPSIDES
PN-1210011A/E	FULL ELECTRIC INTELLIGENT COMPLETION (CI-ELECTRICAL PANELS)	P2C	PN-1210011A/E	TOPSIDES
PN-1223001	FLOW METERING SYSTEM PANEL	P2S	PN-1223001	TOPSIDES
PN-1358501 and PN-1358504	HULL STRUCTURAL TANKS LEVEL, INTERFACE, PRESSURE AND TEMPERATURE MONITORING SYSTEMS	P2S	PN-1358501 and PN-1358504	HULL
PN-5412001	FLARE GAS RECOVERY SYSTEM RELIEF PANEL	P2	PN-5412001	TOPSIDES
PN-5521501	METOCEAN (ENV) & HMS PANEL	P2S	PN-5521501	HULL
PN-5524001A/B	SUBSEA SIGNAL ACQUISITION SYSTEM RACK (SAS)	P2	PN-5524001A/B	TOPSIDES
PN-5529002	RYGID & HYBRID RISER MONITORING SYSTEM - RHMS CABINET	P2	PN-5529002	TOPSIDES
PN-5537501	PRS - POSITIONING	P2S	PN-5537501	HULL

PACKAGE TAG	PACKAGE NAME	TYPE	UCP TAG	LOCATION
	REFERENCE SYSTEM			
PN-5537502	POSITIONING SYSTEM FOR OFFSET DIAGRAM PANEL	P2S	PN-5537502	HULL
SAO-5330501	OILY WATER SEPARATOR PACKAGE	P1	PN-SAO-5330501	HULL
SC-5133501A/B	DIESEL OIL PURIFIER	P1	PN-SC-5133501A/B	HULL
TA-5412001	FLARE IGNITION AND MONITORING PANEL	P2	PN-TA-5412001-01	TOPSIDES
TA-5412001	FLARE TURNDOWN CONTROL SYSTEM	P2	PN-TA-5412001-02	TOPSIDES
TG-5147001A/D	MAIN TURBOGENERATOR UNIT (including WHRU)	P2S	PN-TG-5147001A/D-01	TOPSIDES
UB-5420501A/C	DIESEL HYDRAULIC FIRE WATER PUMPING UNIT	P2	PN-UB-5420501A/C	HULL
UC-1225001A/B	VAPOR RECOVERY UNIT	P2S	PN-UC-1225001A/B-01	TOPSIDES
UC-1231001A/C	MAIN GAS COMPRESSION UNIT	P2S	PN-UC-1231001A/C-01	TOPSIDES
UC-1231002A/C	EXPORTATION GAS COMPRESSION UNIT	P2S	PN-UC-1231002A/C-01	TOPSIDES
UC-1233001A/B	MOLECULAR SIEVE REGENERATION GAS COMPRESSION UNIT	P2	PN-UC-1233001A/B-01	TOPSIDES
UC-1252001A/B	INJECTION GAS COMPRESSION UNIT	P2S	PN-UC-1252001A/B-01	TOPSIDES
UC-1254001A/B	CO2 COMPRESSION UNIT	P2S	PN-UC-1254001A/B-01	TOPSIDES
UC-5134501A/E	INSTRUMENT / SERVICE AIR COMPRESSION UNIT	P2	PN-UC-5134501A/E-01	HULL
UC-5138501	AUXILIARY GENERATOR START UP AIR COMPRESSOR UNIT	P1	PN-UC-5138501	HULL
UC-5412001	FLARE GAS RECOVERY COMPRESSION UNIT	P2	PN-UC-5412001	TOPSIDES
UC-UG-5261501	EMERGENCY GENERATOR START UP AIR COMPRESSOR UNIT	P2	PN-UG-5261501-03	HULL
UD-5122001A/D	FRESH WATER MAKER	P1	PN-UD-5122001A/D	TOPSIDES
UD-5122002A/B	FRESH WATER MAKER FOR OIL DILUTION	P1	PN-UD-5122002A/B-01	TOPSIDES
UE-5121501	SEA WATER ELECTROCHLORINATION UNIT	P1	PN-UE-5121501	HULL
UG-5262502A/B	HULL GENERATOR PACKAGE	P2	PN-UG-5262502A/B-01	HULL
UH-1210001	HYDRAULIC POWER UNIT FOR SUBSEA SYSTEMS	P2	PN-UH-1210001	TOPSIDES
UH-1359506A/B	OFFLOADING HYDRAULIC POWER UNIT	P2	PN-UH-1359506A/B-01	HULL
UH-5139001	HPU FOR TOPSIDES VALVES	P2	PN-UH-5139001	TOPSIDES
UH-5139501	HYDRAULIC VALVES REMOCON UNIT (HULL SYSTEMS)	P2	PN-UH-5139501	HULL
UH-5139502A/B	HYDRAULIC POWER UNIT FOR SUBMERGED PUMPS	P2S	PN-UH-5139502A/B-01	HULL
UH-5139503A/B	AUXILIARY HYDRAULIC UNIT FOR SUBMERGED PUMPS	P2S	PN-UH-5139502A/B-01	HULL
UH-5139504	HYDRAULIC OIL TRANSFER UNIT FOR SUBMERGED PUMPS	P2S	PN-UH-5139502A/B-01	HULL

PACKAGE TAG	PACKAGE NAME	TYPE	UCP TAG	LOCATION
UH-5139505A/B	HYDRAULIC POWER UNIT FOR MOORING SYSTEM (AFT)/(FWD)	P1	PN-UH-5139505A/B	HULL
UH-5139507	HYDRAULIC POWER UNIT FOR MAIN PULL-IN & SPOOLING WINCHES	P1	PN-UH-5139507	HULL
UH-51395078	HYDRAULIC POWER UNIT FOR RISER PULL-IN DECK WINCH	P1	PN-UH-51395078	HULL
UM-5121501A/B	POTABLE WATER MAKER	P1	PN-UM-5121501A/B	HULL
UQ-1261001	OIL AND GAS CHEMICAL INJECTION UNIT	P0	---	TOPSIDES
UQ-1262001	PRODUCED WATER CHEMICAL INJECTION UNIT	P0	---	TOPSIDES
UQ-1263001	INJECTION WATER CHEMICAL INJECTION UNIT	P0	---	TOPSIDES
UR-5252501A/D	WATER CHILLER	P2	PN-UR-5252501A/D	HULL
UT-1233001	MOLECULAR SIEVE UNIT	P2C	PN-UT-1233001	TOPSIDES
UT-1235001	CO2 REMOVAL UNIT	P2C	PN-UT-1235001	TOPSIDES
UT-1238001	HYDROCARBON DEW POINT CONTROL SYSTEM	P2	PN-UT-1238001	TOPSIDES
UT-1251001	ULTRAFILTRATION UNIT	P2	PN-UT-1251001	TOPSIDES
UT-1251002	SULPHATE REMOVAL UNIT	P2	PN-UT-1251002	TOPSIDES
V-5115501	FRESH WATER HYDROPHORE VESSEL	P1	PN-V-5115501	HULL
V-5125501A/B	CALORIFIER WATER VESSEL	P1	PN-V-5125501A/B	HULL
Z-1227001	TEG REGENERATION PACKAGE	P2C	PN-Z-1227001	TOPSIDES
Z-1234001	AMINE REGENERATION PACKAGE	P2	PN-Z-1234001	TOPSIDES
Z-5100501A/B	DIESEL OIL AND FRESH WATER HOSE REELS	P1	PN-Z-5100501A/B	HULL
Z-5121501A/B	ULTRAVIOLET STERILIZER UNIT	P1	PN-Z-5121501A/B	HULL
Z-5122001	FRESH WATER CHLORINATION UNIT	P1	PN-Z-5122001	TOPSIDES
Z-5133502	REFUELING DISPENSER SKID	P1	PN-Z-5133502	HULL
Z-5133503	REFUELING TANKS SKID	P1	PN-Z-5133502	HULL
Z-5133504	REFUELING DRAIN TANK	P1	PN-Z-5133502	HULL
Z-5241001A/B	NITROGEN GENERATION UNIT	P1	PN-Z-5241001A/B	TOPSIDES
Z-5241002A/B	NITROGEN GENERATION UNIT FOR FLARE	P1	PN-Z-5241001A/B	TOPSIDES
Z-5260501A/B	MARINE GROWING PROTECTION SYSTEM	P2	PN-Z-5260501A/B	HULL
Z-5312502A/B	SEWAGE TREATMENT UNIT	P1	PN-Z-5312502A/B	HULL
Z-UT-1251003	VACUUM SYSTEM	P2	PN-Z-UT-1251003	TOPSIDES